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Comprehensive Report

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Pd-Catalyzed 2,3-Allenylation of Oxindoles with 2,3-Allenylic Carbonates

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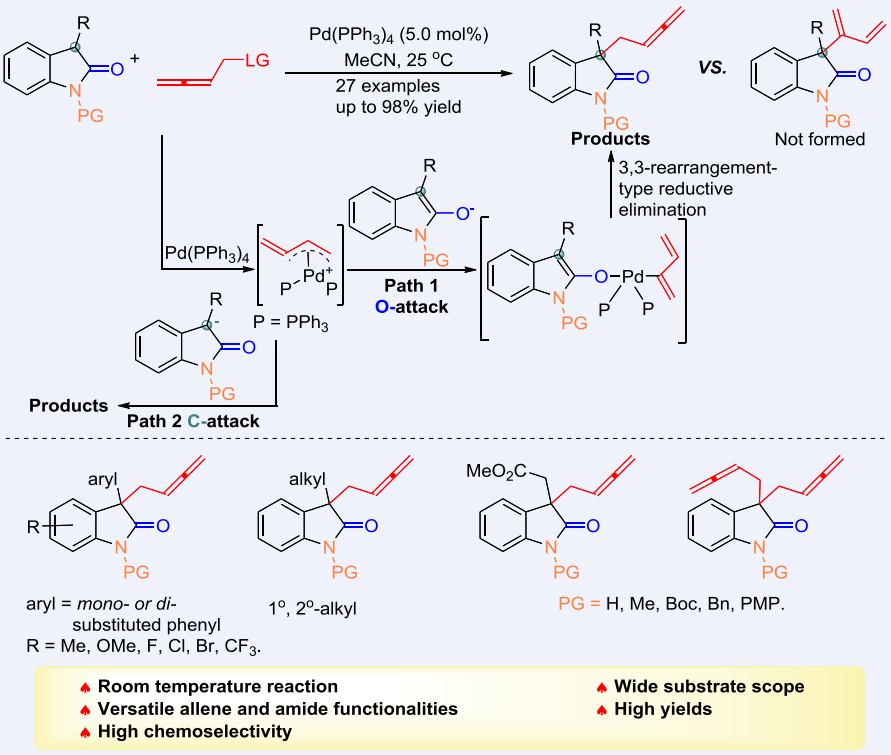
Keywords

Palladium | Allenes | Chemoselectivity | Allenylation | Oxindoles

Main observation and conclusion

Allenes and oxindoles are two classes of very important compounds for medicinal chemistry and organic chemistry. Thus, it is of high interest to combine an allene and an oxindole into one molecule. Herein, the first example of palladium-catalyzed exclusive 2,3-allenylation reaction of oxindoles with 2,3-allenylic carbonates has been successfully developed. A rationale for the selectivity of 2,3-allenylation over the expected 1,3-alkadienylation has been proposed.

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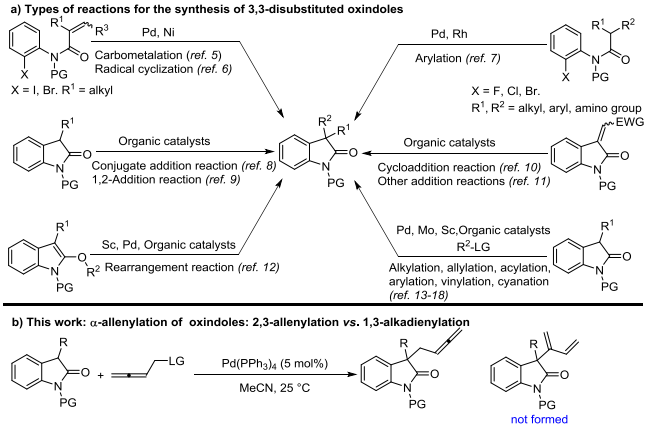
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Background and Originality Content

Oxindoles are a very important class of compounds[1] existing in many natural products such as Horsfiline[2a] and Coerulescine[2b] and various bioactive compounds.[2c-2g] In addition, they are also important building blocks in organic synthesis.[1b,3] Thus, much attention has been paid to the synthesis of this type of compounds[4-18] (Scheme 1a). On the other hand, alleneS[19-20] have become more and more important in medicinal chemistry and organic chemistry due to their unique pharmacological, biological,[21] and chemical properties.[22-23] We reasoned that the 2,3-allenylation of oxindoles would not only be an excellent combination for medicinal chemistry but also dramatically increase the diversity of oxindole derivatives. The challenge is the possible formation of the 1,3-dienylation products due to the nature of oxindole anion (pK_a = 23.5; ~16 for malonate)[24] since nucleophiles with a pK_a value of approximately 20 to 25 are in the border between "hard" and "soft".[25-26] Herein, we wish to report the Pd-catalyzed 2,3-allenylation reaction of oxindoles with 2,3-allenylic carbonates providing oxindoles bearing an allene unit (Scheme 1b).[27]

Scheme 1 Types of reactions for the synthesis of 3,3-disubstituted oxindoles, pK_a value of some compounds, and 2,3-allenylation of oxindoles

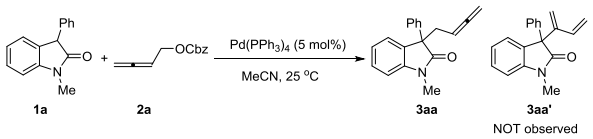


Results and Discussion

Oxindole **1a** and 2,3-allenylic benzyl carbonate **2a** were selected as the model substrates for optimizing reaction conditions. After some screening, the standard reaction conditions have been defined as using $Pd(PPh_3)_4$ (5 mol%) in MeCN at 25 °C, affording the 2,3-allenylation product **3aa** in 91% NMR yield, exclusively. $Pd(PPh_3)_2Cl_2$ or $Pd(OAc)_2$-PPh_3 failed to catalyze the reaction (Table 1, entries 2 & 3). Other palladium(0) sources, such as $Pd(dba)_2$, $Pd_2(dba)_3$, $Pd_2(dba)_3{\cdot}CHCl_3$, and $Pd(dmdba)_2$, were also screened and $Pd(PPh_3)_4$ was still the optimal (Table 1, entries 4—7). With $Pd(PPh_3)_4$ as the catalyst, the reaction in toluene at 25 °C afforded the product **3aa** in 88% yield (Table 1, entry 8). When using etheric solvents such as THF and dioxane, the yield was lower (Table 1, entries 9 & 10). Dichloromethane (DCM), 1,2-dichloroethane (DCE), acetone, ethyl acetate, and ethanol were also inspected with no better result (Table 1, entries 11—15). It is worth noting that **3aa** was selectively generated in this reaction, and 1,3-dienylation product **3aa'** was NOT observed in all the cases during our optimization of the reaction conditions.

With the optimized reaction conditions (Table 1, entry 1), the scope of the *N*-substituent was investigated. Methyl or *tert*-butoxycarbonyl groups provided the products **3aa** and **3ba** in 90% yield and 88% yield, respectively (Table 2, entries 1 & 2). The

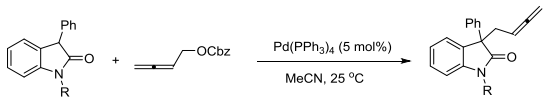
Table 1 Optimization of reaction conditions for Pd-catalyzed 2,3-allenylation reaction of oxindole **1a** with **2a**[a]



Entry	Deviation from standard conditions	NMR yields[b]/% (**3aa**/**1a**)
1	Standard conditions	91/0
2[c,d]	$Pd(OAc)_2$ (5 mol%) as catalyst	0/72
3[d]	$Pd(PPh_3)_2Cl_2$ as catalyst	0/98
4[c]	$Pd(dba)_2$ (5 mol%) as catalyst	84/0
5[c]	$Pd_2(dba)_3$ (2.5 mol%) as catalyst	83/0
6[c]	$Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%) as catalyst	79/0
7[c]	$Pd(dmdba)_2$ (5 mol%) as catalyst	83/0
8	Toluene as solvent	88/0
9	THF as solvent	76/0
10	Dioxane as solvent	88/0
11	DCM as solvent	90/0
12	DCE as solvent	88/0
13	Acetone as solvent	68/0
14	Ethyl acetate as solvent	70/0
15	Ethanol as solvent	77/0

[a] Standard reaction conditions: The reaction of **1a** (0.24 mmol), **2a** (0.20 mmol) and $Pd(PPh_3)_4$ (5 mol%) in MeCN (1.0 mL) was conducted at 25 °C. [b] NMR yield with CH_2Br_2 as the internal standard. [c] PPh_3 (10 mol%) was added. [d] DCE as solvent.

Table 2 Survey on *N*-substituents[a]



Entry	R	Yield[b]/% (**3**)
1	Me (**1a**)	90 (**3aa**)
2	Boc (**1b**)	88 (**3ba**)
3	Bn (**1c**)	80 (**3ca**)
4	PMP (**1c'**)	84 (**3c'a**)
5	H (**1d**)	81 (**3da**)

[a] Standard reaction conditions: The reaction of **1** (1.2 mmol), **2a** (1.0 mmol) and $Pd(PPh_3)_4$ (5 mol%) in MeCN (5.0 mL) was conducted at 25 °C. [b] Isolated yield.

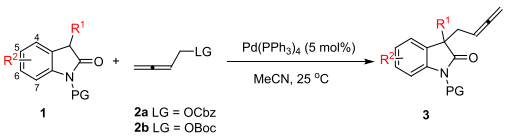
reaction also worked with easily removable benzyl group-substituted oxindole **1c**, *p*-methoxyphenyl group-substituted oxindole **1c'** (with the reported pK_a of 23.5),[24] and even the *N*-non-substituted oxindole **1d**, leaving further opportunities for elaboration of the *N*-substituent (Table 2, entries 3—5). Again, the corresponding **3aa'**-type 1,3-diene products were NOT formed (see Scheme 1b).

A series of 3-mono-substituted oxindoles smoothly underwent this transformation, producing a variety of 3-(2,3-butadienyl) oxindoles in moderate to good yields. 3-Aryl-substituted oxindoles bearing different substituents on the 3-phenyl were tested. As shown in Table 3, the variation of electronic and steric effects was tolerated: the substrates with methoxy, fluoro, chloro, methyl, ethyl, isopropyl at the *para*- or *ortho*-position of the phenyl ring could react with 2,3-allenylic carbonate **2** to afford the corresponding products in 74%—99% yields (Table 3, entries 3—9). 2,4-Dimethyl, 2,5-dimethyl, 4-methoxy-2-methyl, and 4-chloro-2-methyl substituted aryl groups were also tolerated (66%—96%, Table 3, entries 10—13). Further exploration of the substrate

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Table 3 Substrate scope of $Pd(PPh_3)_4$-catalyzed 2,3-allenylation of oxindoles[a]



Entry	R^1	R^2	PG	Yield[b]/% (3)
1[c]	C_6H_5 (**1a**)	H	Me	90 (**3aa**)
2[c]	C_6H_5 (**1b**)	H	Boc	88 (**3ba**)
3[c]	4-$MeOC_6H_4$ (**1e**)	H	Me	89 (**3eb**)
4[c]	4-FC_6H_4 (**1f**)	H	Me	87 (**3fa**)
5[c]	4-ClC_6H_4 (**1g**)	H	Me	99 (**3ga**)
6[c]	2-MeC_6H_4 (**1h**)	H	Boc	98 (**3hb**)
7	2-EtC_6H_4 (**1i**)	H	Boc	84 (**3ib**)
8	2-$MeOC_6H_4$ (**1j**)	H	Boc	78 (**3jb**)
9	2-*i*-PrC_6H_4 (**1k**)	H	Boc	74 (**3kb**)
10	2-Me-4-MeC_6H_3 (**1l**)	H	Boc	66 (**3lb**)
11	2-Me-5-MeC_6H_3 (**1m**)	H	Boc	77 (**3mb**)
12	4-MeO-2-MeC_6H_3 (**1n**)	H	Boc	80 (**3nb**)
13	4-Cl-2-MeC_6H_3 (**1o**)	H	Boc	96 (**3ob**)
14[c]	2-MeC_6H_4 (**1p**)	5-Me	Boc	80 (**3pb**)
15	2-MeC_6H_4 (**1q**)	5-MeO	Boc	58 (**3qb**)
16[d]	2-MeC_6H_4 (**1r**)	5-F	Boc	95 (**3rb**)
17[c]	2-MeC_6H_4 (**1s**)	5-Cl	Boc	94 (**3sb**)
18[c]	2-MeC_6H_4 (**1t**)	5-Br	Boc	95 (**3tb**)
19[c]	C_6H_5 (**1u**)	6-Cl	Me	96 (**3ua**)
20[c]	2-MeC_6H_4 (**1v**)	6-Cl	Boc	89 (**3vb**)
21	2-MeC_6H_4 (**1w**)	6-Br	Boc	44 (**3wb**)
22	2-MeC_6H_4 (**1x**)	7-Cl	Boc	92 (**3xb**)
23	2-MeC_6H_4 (**1y**)	7-CF_3	Boc	85 (**3yb**)
24	*n*-C_7H_{15} (**1z**)	H	Boc	65 (**3zb**)
25	CH_2CO_2Me (**1A**)	H	Me	60 (**3Ab**)
26	*i*-Pr (**1B**)	H	Boc	70 (**3Bb**)
27	Cy (**1C**)	H	Boc	69 (**3Cb**)

[a] Standard reaction conditions: The reaction of **1** (0.6 mmol), **2b** (0.5 mmol) and $Pd(PPh_3)_4$ (5 mol%) in MeCN (2.5 mL) was conducted at 25 °C. [b] Isolated yield. [c] The reaction of **1** (1.2 mmol), **2** (1.0 mmol) and $Pd(PPh_3)_4$ (5 mol%) in MeCN (5.0 mL) was conducted at 25 °C. [d] The reaction of **1r** (0.5 mmol), **2b** (0.6 mmol) and $Pd(PPh_3)_4$ (5 mol%) in MeCN (2.5 mL) was conducted at 25 °C.

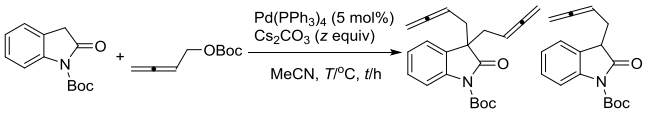
scope was focused on the left hand phenyl ring on the oxindoles. Various 3-(*o*-tolyl)-oxindoles substituted with electron-donating groups (methyl and methoxy) and electron-withdrawing groups (fluoro, chloro, bromo and trifluoromethyl) also reacted to afford the 2,3-allenylation products (44%—96%, Table 3, entries 14—23). In addition, this reaction also worked with 3-alkyl-substituted oxindoles affording the corresponding products in 60%—70% (Table 3, entries 24—27).

Besides, 3-non-substituted oxindole **1D** and 2,3-allenylic *tert*-butyl carbonate **2b** also worked well in this reaction, and only double-substituted product **3Db** was formed and no single-substituted product **3Db'** was observed (Table 4, entry 1). Even if the amount of **1D** was 2.3 equiv, the reaction at 25 °C in the absence of a base still afforded double 2,3-allenylation product **3Db** albeit in a low yield (Table 4, entry 3).

However, the reaction of 1,1-disubstituted allenylic carbonate as the substrates failed (**2c** & **2d**) (Scheme 2, eqs (1) & (2)). No reaction occurred to pyrrolidin-2-one (**1E**) and 1-methylpyrrolidin-2-one (**1F**) (Scheme 2, eqs (3) & (4)).

As mentioned in the introduction, the regioselectivity of the Pd-catalyzed allenol derivatives with nucleophiles depended on the nature of the nucleophiles: Gore[28a] pioneered the Pd-catalyzed 2,3-allenylation reaction of 2,3-allenol derivatives with soft

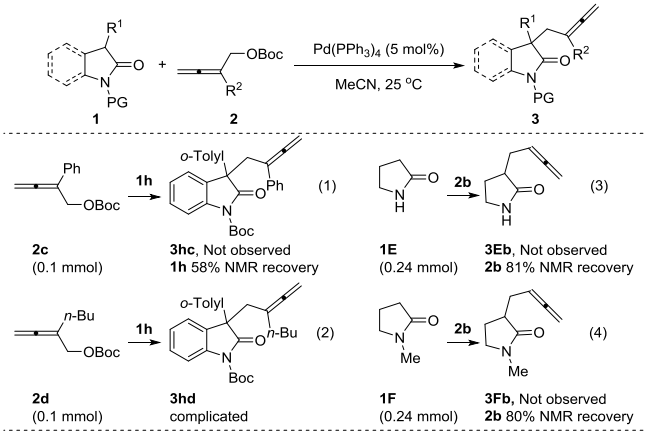
Table 4 $Pd(PPh_3)_4$-catalyzed 2,3-allenylation of 3-non-substituted oxindole **1D**[a]



Entry	**1D** (*x* mmol)	**2b** (*y* mmol)	Cs_2CO_3 (*z* equiv)	*T*/°C	*t*/h	**3Db** NMR yield/% (**3Db**/**3Db'**)	**3Db'** NMR recovery/% (**1D**)
1	0.5	1.2	1.2	50	18	78 (76)/0	0
2	0.23	0.1	1.2	25	12	37/0	43
3	0.23	0.1	0	25	12	21/0	73
4	0.24	0.1	1.2	50	18	16/0	33
5	0.24	0.1	0	50	18	30/0	42

[a] Reaction conditions: The reaction of **1D**, **2b**, Cs_2CO_3, and $Pd(PPh_3)_4$ (5 mol%) in MeCN was conducted. NMR yield with CH_2Br_2 as the internal standard. Isolated yield is given in the parentheses.

Scheme 2 Some failed substrates of $Pd(PPh_3)_4$-catalyzed 2,3-allenylation reaction[a]



[a] Reaction conditions: The reaction of **1** (1.2 equiv), **2**, and $Pd(PPh_3)_4$ (5 mol%) in MeCN was conducted. NMR yield or recovery with CH_2Br_2 as the internal standard.

nucleophile forming allene products, exclusively[28-29] (Scheme 3a, path a). On the other hand, Vermeer[30a] reported the Pd-catalyzed coupling reaction of 2,3-allenol derivatives with hard nucleophile, affording 1,3-diene product (Scheme 3a, path b). Due to the nucleophilic nature of oxindole anion, 1,3-dienes should be the expected products,[30] however, 2,3-allenylation products were formed exclusively here. We further performed a deuterium-labeling experiment to gain some insights into the reaction mechanism. The reaction of oxindole **1h** with terminal fully deuteriated 2,3-allenylic *tert*-butyl carbonate **2b-*d*$_2$** gave product **3hb-*d*$_2$** (Scheme 3b). Thus, we proposed a possible mechanism as shown in Scheme 3c. Firstly, oxidative addition of **2b** with Pd(0) would form η^1-dienyl-Pd, which would immediately isomerize to the delocalized methylene-π-allyl palladium η^3-**int I**[31] *via* the σ-π rearrangement involving the coordination of the terminal C=C bond with Pd (Scheme 3c). The oxindole enol anion may approach the methylene-π-allyl palladium intermediate by the direct oxygen attack of the enolate at the Pd atom to form **int II**, which may most probably proceed *via* the inner-sphere 3,3-rearrangement-type reductive elimination[32] to afford **3**. It is also difficult to exclude the possibility of having η^3-**int I** reacted directly with the oxindole anion to form the final product **3**.

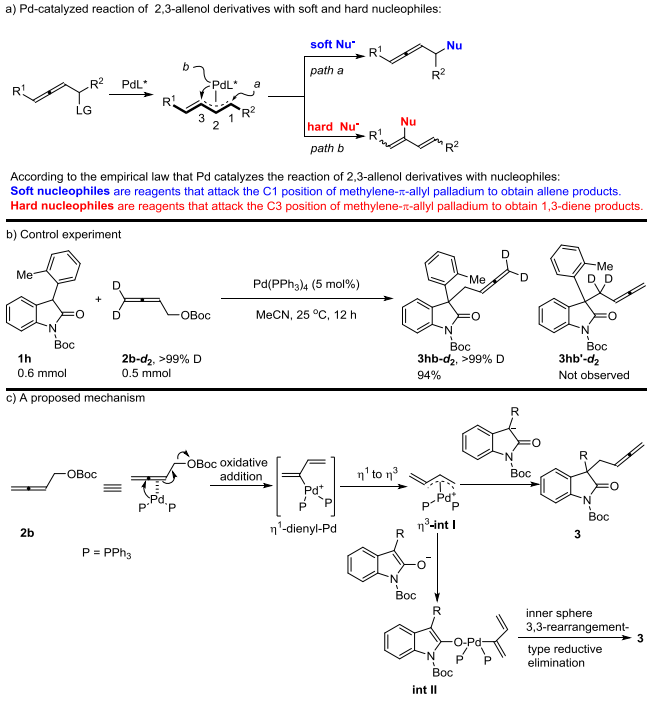
Conclusions

In conclusion, we have developed the first example of Pd-catalyzed 2,3-allenylation reaction of oxindoles with 2,3-

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Scheme 3 Pd-catalyzed reaction of 2,3-allenol derivatives with soft and hard nucleophiles, control experiment, and a proposed mechanism



allenylic carbonates for the efficient construction of allene derivatives bearing an oxindole unit. The reaction enjoys a wide substrate range and good yields with a different mechanism. Further studies are being pursued in our laboratory.

Experimental

Typical procedure for the synthesis of 3-(2,3-butadienyl)-1-methyl-3-phenylindolin-2-one (3aa). To a flame-dried Schlenk tube were added $Pd(PPh_3)_4$ (57.5 mg, 0.05 mmol), **1a** (266.5 mg, 1.2 mmol), and **2a** (203.8 mg, 1.0 mmol)/MeCN (5.0 mL) sequentially under Ar atmosphere at room temperature. The resulting mixture was stirred at 25 °C for 5.5 h as monitored by TLC and filtrated through a short column of silica gel (height: 3 cm, *Φ*: 3.5 cm) eluted with diethyl ether (60 mL). After evaporation, the residue was purified by column chromatography on silica gel to afford **3aa** (246.7 mg, 90%) as an oil [eluent: petroleum ether/ethyl acetate = 15/1]. ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.37 (d, *J* = 7.6 Hz, 2H, ArH), 7.36—7.18 (m, 5H, ArH), 7.10 (t, *J* = 7.4 Hz, 1H, ArH), 6.87 (d, *J* = 7.6 Hz, 1H, ArH), 4.80—4.70 (quintet, *J* = 7.0 Hz, 1H, =CH), 4.56—4.48 (m, 1H, one proton of $=CH_2$), 4.44—4.36 (m, 1H, one proton of $=CH_2$), 3.18 (s, 3H, CH_3), 3.03—2.90 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 210.0, 177.7, 143.9, 139.2, 131.4, 128.4, 128.3, 127.3, 127.0, 125.2, 122.4, 108.2, 84.3, 74.4, 56.3, 37.0, 26.3; IR (neat) *ν*: 3056, 2930, 1954, 1707, 1608, 1491, 1468, 1369, 1344, 1252, 1182, 1158, 1128, 1086, 1019 cm^{-1}; MS (70 eV, EI) *m*/*z* (%): 275 [M$^+$, 46.27], 222 (100); HRMS Calcd for $C_{19}H_{17}NO$ [M$^+$]: 275.1310; Found: 275.1308.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202100402.

Acknowledgement

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References

[1] For selected reviews on natural products and bioactive compounds containing oxindole scaffolds, see: (a) Marti, C.; Carreira, E. M. Construction of Spiro[pyrrolidine-3,3'-oxindoles]-Recent Applications to the Synthesis of Oxindole Alkaloids. *Eur. J. Org. Chem.* **2003**, 2209–2219; (b) Galliford, C. V.; Scheidt, K. A. Pyrroldinyl-Spirooxindole Natural Products as Inspirations for the Development of Potential Therapeutic Agents. *Angew. Chem. Int. Ed.* **2007**, *46*, 8748–8758; (c) Peddibhotla, S. 3-Substituted-3-hydroxy-2-oxindole, an Emerging New Scaffold for Drug Discovery with Potential Anti-Cancer and other Biological Activities. *Curr. Bioact. Compd.* **2009**, *5*, 20–38; (d) Millemaggi, A.; Taylor, R. J. K. 3-Alkenyl-oxindoles: Natural Products, Pharmaceuticals, and Recent Synthetic Advances in Tandem/Telescoped Approaches. *Eur. J. Org. Chem.* **2010**, 4527–4547.

[2] (a) Jossang, A.; Jossang, P.; Hadi, H. A.; Sévenet, T.; Bodo, B. Horsfiline, an Oxindole Alkaloid from Horsfieldia superba. *J. Org. Chem.* **1991**, *56*, 6527–6530; (b) Anderton, N.; Cockrum, P. A.; Colegate, S. M.; Edgar, J. A.; Flower, K.; Vit, I.; Willing, R. I. Oxindoles from Phalaris Coerulescens. *Phytochemistry* **1998**, *48*, 437–439; (c) Canas-Rodriguez, A.; Leeming, P. R. *N*-Phenyllndoline Derivatives-a New Class of Antidepressant Agents. *Nature* **1969**, *223*, 75–76; (d) Herth, M. M.; Andersen, V. L.; Hansen, H. D.; Stroth, N.; Volk, B.; Lehel, S.; Dyssegaard, A.; Ettrup, A.; Svenningsson, P.; Knudsen, G. M.; Kristensen, J. L. Evaluation of 3-Ethyl-3-(phenylpiperazinylbutyl)oxindoles as PET Ligands for the Serotonin 5-HT$_7$ Receptor: Synthesis, Pharmacology, Radiolabeling, and *in Vivo* Brain Imaging in Pigs. *J. Med. Chem.* **2015**, *58*, 3631–3636; (e) Grese, T. A.; Jadhav, P. K.; Neel, D. A.; Steinberg, M. I.; Lander, P. A. Dihydroindol-2-one derivatives as steroid hormone nuclear receptor modulators. *US20050054712-A1*, **2005**; (f) Halperin, J. A.; Natarajan, A.; Aktas, H.; Fan, Y.-H.; Chen, H. 3-3-Di-substituted-oxindoles as inhibitors of translation initation. *US20070099976-A1*, **2007**; (g) Luk, K.-C.; So, S.-S.; Zhang, J.; Zhang, Z. Oxindole derivatives. *US20060293319-A1*, **2006**.

[3] For selected reviews on reactions of oxindoles, see: (a) Sumpter, W. C. The Chemistry of Oxindole. *Chem. Rev.* **1945**, *37*, 443–479. For selected reports on the total synthesis of oxindole as a synthesis intermediate, see: (b) Onishi, T.; Sebahar, P. R.; Williams, R. M. Concise, Asymmetric Total Synthesis of Spirotryprostatin A. *Org. Lett.* **2003**, *5*, 3135–3137; (c) Meyers, C.; Carreira, E. M. Total Synthesis of (–)-Spirotryprostatin B. *Angew. Chem. Int. Ed.* **2003**, *42*, 694–696; (d) Adhikari, S.; Caille, S.; Hanbauer, M.; Ngo, V. X.; Overman, L. E. Asymmetric Construction of Quaternary Carbon Stereocenters: High Stereoselection in Mukaiyama Aldol Reactions of 2-Siloxyindoles with Chiral Aldehydes. *Org. Lett.* **2005**, *7*, 2795–2797; (e) Kumar, N.; Das, M. K.; Ghosh, S.; Bisai, A. Development of catalytic deacylative alkylations (DaA) of 3-acyl-2-oxindoles: total synthesis of meso-chimonanthine and related alkaloids. *Chem. Commun.* **2017**, *53*, 2170–2173.

[4] For selected reviews on synthesis of 3,3-disubstituted oxindoles, see: (a) Zhou, F.; Liu, Y.-L.; Zhou, J. Catalytic Asymmetric Synthesis of Oxindoles Bearing a Tetrasubstituted Stereocenter at the C-3 Position. *Adv. Synth. Catal.* **2010**, *352*, 1381–1407; (b) Singh, G. S.; Desta, Z. Y. Isatins as Privileged Molecules in Design and Synthesis of SpiroFused Cyclic Frameworks. *Chem. Rev.* **2012**, *112*, 6104–6155; (c) Shen, K.; Liu, X.; Lin, L.; Feng, X. Recent progress in enantioselective synthesis of C3-functionalized oxindoles: rare earth metals take action. *Chem. Sci.* **2012**, *3*, 327–334; (d) Cheng, D.; Ishihara, Y.; Tan, B.; Barbas, III, C. F. Organocatalytic Asymmetric Assembly Reactions: Synthesis of Spirooxindoles *via* Organocascade Strategies. *ACS Catal.* **2014**, *4*, 743–762; (e) Cao, Z.-Y.; Zhou, F.; Zhou, J. Development of Synthetic Methodologies *via* Catalytic Enantioselective Synthesis of

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16147065, 2021, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.202100402 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3,3-Disubstituted Oxindoles. *Acc. Chem. Res.* **2018**, *51*, 1443–1454.

[5] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* Pd-catalyzed carbometalation reactions, see: (a) Ashimori, A.; Overman, L. E. Catalytic Asymmetric Synthesis of Quarternary Carbon Centers. Palladium-Catalyzed Formation of Either Enantiomer of Spirooxindoles and Related Spirocyclics Using a Single Enantiomer of a Chiral Diphosphine Ligand. *J. Org. Chem.* **1992**, *57*, 4571–4572; (b) Busacca, C. A.; Grossbach, D.; So, R. C.; O'Brien, E. M.; Spinelli, E. M. Probing Electronic Effects in the Asymmetric Heck Reaction with the BIPI Ligands. *Org. Lett.* **2003**, *5*, 595–598; (c) Pinto, A.; Jia, Y.; Neuville, L.; Zhu, J. Palladium-Catalyzed Enantioselective Domino Heck Cyanation Sequence: Development and Application to the Total Synthesis of Esermethole and Physostigmine. *Chem.-Eur. J.* **2007**, *13*, 961–967; (d) Yasui, Y.; Kamisaki, H.; Takemoto, Y. Enantioselective Synthesis of 3,3-Disubstituted Oxindoles through Pd-Catalyzed Cyanoamidation. *Org. Lett.* **2008**, *10*, 3303–3306; For selected reports on synthesis of 3,3-disubstituted oxindoles *via* Ni-catalyzed carbometalation reactions, see: (e) Desrosiers, J.-N.; Hie, L.; Biswas, S.; Zatolochnaya, O. V.; Rodriguez, S.; Lee, H.; Grinberg, N.; Haddad, N.; Yee, N. K.; Garg, N. K.; Senanayake, C. H. Construction of Quaternary Stereocenters by Nickel-Catalyzed Heck Cyclization Reactions. *Angew. Chem. Int. Ed.* **2016**, *55*, 11921–11924; (f) Desrosiers, J.-N.; Wen, J.; Tcyrulnikov, S.; Biswas, S.; Qu, B.; Hie, L.; Kurouski, D.; Wu, L.; Grinberg, N.; Haddad, N.; Busacca, C. A.; Yee, N. K.; Song, J. J.; Garg, N. K.; Zhang, X.; Kozlowski, M. C.; Senanayake, C. H. Enantioselective Nickel-Catalyzed Mizoroki-Heck Cyclizations to Generate Quaternary Stereocenters. *Org. Lett.* **2017**, *19*, 3338–3341; (g) Ma, T.; Chen, Y.; Li, Y.; Ping, Y.; Kong, W. Nickel-Catalyzed Enantioselective Reductive Aryl Fluoroalkenylation of Alkenes. *ACS Catal.* **2019**, *9*, 9127–9133.

[6] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* radical cyclisation reactions, see: (a) Jones, K.; Thompson, M.; Wright, C. An Efficient Synthesis of Spiro[cyclohexane-1,3'-indol-2'(3'*H*)-ones] *via* Radical Cyclisation. *J. Chem. Soc., Chem. Commun.* **1986**, 115–116; (b) Bowman, W. R.; Heaney, H.; Jordan, B. M. Synthesis of Oxindoles by Radical Cyclisation. *Tetrahedron Lett.* **1988**, *29*, 6657–6660; (c) Jones, K.; Wilkinson, J.; Ewin, R. Intramolecular Reactions Using Amide Links: Aryl Radical Cyclisation of Silylated Acryloylanilides. *Tetrahedron Lett.* **1994**, *35*, 7673–7676; (d) Cabri, W.; Candiani, I.; Colombo, M.; Franzoi, L.; Bedeschi, A. Non-Toxic Ligands in Samarium Diiodide-Mediated Cyclizations. *Tetrahedron Lett.* **1995**, *36*, 949–952.

[7] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* Intramolecular arylation reactions, see: (a) Shaughnessy, K. H.; Hamann, B. C.; Hartwig, J. F. Palladium-Catalyzed Inter- and Intramolecular α-Arylation of Amides. Application of Intramolecular Amide Arylation to the Synthesis of Oxindoles. *J. Org. Chem.* **1998**, *63,* 6546–6553; (b) Glorius, F.; Altenhoff, G.; Goddard, R.; Lehmann, C. Oxazolines as chiral building blocks for imidazolium salts and N-heterocyclic carbene ligands. *Chem. Commun.* **2002**, 2704–2705; (c) Arao, T.; Kondo, K.; Aoyama, T. Development of an N-heterocyclic carbene ligand based on concept of chiral mimetic. *Tetrahedron Lett.* **2006**, *47*, 1417–1420; (d) Kündig, E. P.; Seidel, T. M.; Jia, Y.-X.; Bernardinelli, G. Bulky Chiral Carbene Ligands and Their Application in the PalladiumCatalyzed Asymmetric Intramolecular α-Arylation of Amides. *Angew. Chem. Int. Ed.* **2007**, *46*, 8484–8487; (e) Marsden, S. P.; Watson, E. L.; Raw, S. A. Facile and General Synthesis of Quaternary 3-Aminooxindoles. *Org. Lett.* **2008**, *10*, 2905–2908; (f) Liu, L.; Ishida, N.; Ashida, S.; Murakami, M. Synthesis of Chiral *N*-Heterocyclic Carbene Ligands with Rigid Backbones and Application to the Palladium Catalyzed Enantioselective Intrmolecular α-Arylation of Amides. *Org. Lett.* **2011**, *13*, 1666–1669.

[8] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* conjugate addition reactions, see: (a) He, R.; Ding, C.; Maruoka, K. Phosphonium Salts as Chiral Phase-Transfer Catalysts: Asymmetric Michael and Mannich Reactions of 3-Aryloxindoles. *Angew. Chem. Int. Ed.* **2009**, *48*, 4559–4561; (b) Kato, Y.; Furutachi, M.; Chen, Z.; Mitsunuma, H.; Matsunaga, S.; Shibasaki, M. A Homodinuclear $Mn(III)_2$-Schiff Base Complex for Catalytic Asymmetric 1,4-Additions of Oxindoles to Nitroalkenes. *J. Am. Chem. Soc.* **2009**, *131*, 9168–9169; (c) Bui, T.; Syed, S.; Barbas, III, C. F. Thiourea-Catalyzed Highly Enantio- and Diastereoselective Additions of Oxindoles to Nitroolefins: Application to the Formal Synthesis of (+)-Physostigmine. *J. Am. Chem. Soc.* **2009**, *131*, 8758–8759; (d) Galzerano, P.; Bencivenni, G.; Pesciaioli, F.; Mazzanti, A.; Giannichi, B.; Sambri, L.; Bartoli, G.; Melchiorre, P. Asymmetric Iminium Ion Catalysis with a Novel Bifunctional Primary Amine Thiourea: Controlling Adjacent Quaternary and Tertiary Stereocenters. *Chem.-Eur. J.* **2009**, *15*, 7846–7849; (e) Bravo, N.; Mon, I.; Companyó, X.; Alba, A.-N.; Moyano, A.; Rios, R. Enantioselective addition of oxindoles to aliphatic α,β-unsaturated aldehydes. *Tetrahedron Lett.* **2009**, *50*, 6624–6626; (f) Duan, S.-W.; An, J.; Chen, J.-R.; Xiao, W.-J. Facile Synthesis of Enantioenriched C^{γ}-Tetrasubstituted α-Amino Acid Derivatives *via* an Asymmetric Nucleophilic Addition/Protonation Cascade. *Org. Lett.* **2011**, *13*, 2290–2293; (g) Zhang, T.; Cheng, L.; Hameed, S.; Liu, L.; Wang, D.; Chen, Y.-J. Highly enantioselective Michael addition of 2-oxindoles to vinyl selenone in RTILs catalyzed by a Cinchona alkaloid-based thiourea. *Chem. Commun.* **2011**, *47*, 6644–6646; (h) Ding, M.; Zhou, F.; Liu, Y.-L.; Wang, C.-H.; Zhao, X.-L.; Zhou, J. Cinchona alkaloid-based phosphoramide catalyzed highly enantioselective Michael addition of unprotected 3-substituted oxindoles to nitroolefins. *Chem. Sci.* **2011**, *2*, 2035–2039; (i) Mitsunuma, H.; Shibasaki, M.; Kanai, M.; Matsunaga, S. Catalytic Asymmetric Total Synthesis of Chimonanthine, Folicanthine, and Calycanthine through Double Michael Reaction of Bisoxindole. *Angew. Chem. Int. Ed.* **2012**, *51*, 5217–5221; (j) Sun, W.; Zhu, G.; Wu, C.; Hong, L.; Wang, R. "Organo-Metal" Synergistic Catalysis: The 1+1>2 Effect for the Construction of Spirocyclopentene Oxindoles. *Chem.-Eur. J.* **2012**, *18*, 13959–13963; (k) Zong, L.; Du, S.; Chin, K. F.; Wang, C.; Tan, C.-H. Enantioselective Synthesis of Quaternary Carbon Stereocenters: Addition of 3-Substituted Oxindoles to Vinyl Sulfone Catalyzed by Pentanidiums. *Angew. Chem. Int. Ed.* **2015**, *54*, 9390–9393; (l) Wei, Y.; Wen, S.; Liu, Z.; Wu, X.; Zeng, B.; Ye, J. Diastereodivergent Catalytic Asymmetric Michael Addition of 2-Oxindoles to α,β-Unsaturated Ketones by Chiral Diamine Catalysts. *Org. Lett.* **2015**, *17*, 2732–2735; (m) Zhao, K.; Zhi, Y.; Wang, A.; Enders, D. Asymmetric Organocatalytic Synthesis of 3-Diarylmethine Substituted Oxindoles Bearing a Quaternary Stereocenter *via* 1,6-Conjugate Addition to *para*-Quinone Methides. *ACS Catal.* **2016**, *6*, 657–660; (n) Wang, S.; Guo, Z.; Wu, Y.; Liu, W.; Liu, X.; Zhang, S.; Sheng, C. Organocatalytic asymmetric synthesis of highly functionalized spiro-thiazolone-cyclopropane-oxindoles bearing two vicinal spiro quaternary centers. *Org. Chem. Front.* **2019**, *6*, 1442–1447.

[9] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* 1,2-addition reactions, see: (a) Tian, X.; Jiang, K.; Peng, J.; Du, W.; Chen, Y.-C. Organocatalytic Stereoselective Mannich Reaction of 3-Substituted Oxindoles. *Org. Lett.* **2008**, *10*, 3583–3586; (b) Li, J.; Du, T.; Zhang, G.; Peng, Y. 3-Bromooxindoles as nucleophiles in asymmetric organocatalytic Mannich reactions with *N*-Ts-imines. *Chem. Commun.* **2013**, *49*, 1330–1332; (c) Shimizu, S.; Tsubogo, T.; Xu, P.; Kobayashi, S. Calcium-Catalyzed Asymmetric Synthesis of 3-Tetrasubstituted Oxindoles: Efficient Construction of Adjacent Quaternary and Tertiary Chiral Centers. *Org. Lett.* **2015**, *17*, 2006–2009.

[10] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* cycloaddition reactions, see: (a) Sebahar, P. R.; Williams, R. M. The Asymmetric Total Synthesis of (+)- and (–)-Spirotryprostatin B. *J. Am. Chem. Soc.* **2000**, *122*, 5666–5667; (b) Malinakova, H. C.; Liebeskind, L. S. Enantiocontrolled Synthesis of Spirooxindoles Based on the [5+2] Cycloaddition of a $Tp(CO)_2Mo$(pyridinyl) Scaffold (Tp = Hydridotrispyrazolylborate). *Org. Lett.* **2002**, *2*, 4083–4086; (c) Trost, B. M.; Cramer, N.; Bernsmann, H. Concise Total Synthesis of (±)-Marcfortine B. *J. Am. Chem. Soc.* **2007**, *129,* 3086–3087; (d) Chen, X.-H.; Wei, Q.; Luo, S.-W.; Xiao, H.; Gong, L.-Z. Organocatalytic Synthesis of Spiro[pyrrolidin-3,3'-oxindoles] with High Enantiopurity and Structural Diversity. *J. Am. Chem. Soc.* **2009**, *131,* 13819–13825; (e) Voituriez, A.; Pinto, N.; Neel, M.; Retailleau, P.; Marinetti, A. An Organocatalytic [3+2] Cyclisation Strategy for the Highly Enantioselective Synthesis of

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16147065, 2021, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.202100402 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Spirooxindoles. *Chem.-Eur. J.* **2010***, 16,* 12541–12544; (f) Tan, B.; Candeias, N. R.; Barbas, III., C. F. Core-Structure-Motivated Design of a Phosphine-Catalyzed [3+2] Cycloaddition Reaction: Enantioselective Syntheses of Spirocyclopenteneoxindoles. *J. Am. Chem. Soc.* **2011**, *133*, 4672–4675.

[11] Yu, J.-S.; Liao, F.-M.; Gao, W.-M.; Liao, K.; Zuo, R.-L.; Zhou, J. Michael Addition Catalyzed by Chiral Secondary Amine Phosphoramide Using Fluorinated Silyl Enol Ethers: Formation of Quaternary Carbon Stereocenters. *Angew. Chem. Int. Ed.* **2015**, *54*, 7381–7385.

[12] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* rearrangement reactions, see: (a) Hills, I. D.; Fu, G. C. Catalytic Enantioselective Synthesis of Oxindoles and Benzofuranones That Bear a Quaternary Stereocenter. *Angew. Chem. Int. Ed.* **2003**, *42*, 3921–3924; (b) Linton, E. C.; Kozlowski, M. C. Catalytic Enantioselective Meerwein-Eschenmoser Claisen Rearrangement: Asymmetric Synthesis of Allyl Oxindoles. *J. Am. Chem. Soc.* **2008**, *130*, 16162–16163; (c) Cao, T.; Deitch, J.; Linton, E. C.; Kozlowski, M. C. Asymmetric Synthesis of Allenyl Oxindoles and Spirooxindoles by a Catalytic Enantioselective Saucy-Marbet Claisen Rearrangement. *Angew. Chem. Int. Ed.* **2012**, *51*, 2448–2451; (d) Lai, Z.-W.; Liu, C.; Sun, H.; You, S.-L. Asymmetric Synthesis of 3-Allyloxindoles and 3-Allenyloxindoles by Scandium(III)-Catalyzed Claisen Rearrangement Reactions. *Chin. J. Chem.* **2017**, *35*, 1512–1516.

[13] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* alkylation reactions, see: (a) Lee, T. B. K.; Wong, G. S. K. Asymmetric Alkylation of Oxindoles: An Approach to the Total Synthesis of (–)-Physostigmine. *J. Org. Chem.* **1991**, *56*, 872–875; (b) Huang, A.; Kodanko, J. J.; Overman, L. E. Asymmetric Synthesis of Pyrrolidinoindolines. Application for the Practical Total Synthesis of (–)-Phenserine. *J. Am. Chem. Soc.* **2004**, *126*, 14043–14053; (c) Trost, B. M.; Czabaniuk, L. C. Palladium-Catalyzed Asymmetric Benzylation of 3-Aryl Oxindoles. *J. Am. Chem. Soc.* **2010**, *132*, 15534–15536; (d) Ohmatsu, K.; Kiyokawa, M.; Ooi, T. Chiral 1,2,3-Triazoliums as New Cationic Organic Catalysts with Anion-Recognition Ability: Application to Asymmetric Alkylation of Oxindoles. *J. Am. Chem. Soc.* **2011**, *133*, 1307–1309; (e) Zhang, T.; Qiao, Z.; Wang, Y.; Zhong, N.; Liu, L.; Wang, D.; Chen, Y.-J. Asymmetric direct α-alkylation of 2-oxindoles with Michler's hydrol catalyzed by bis-cinchona alkaloid-Brønsted acid *via* an S_N1-type pathway. *Chem. Commun.* **2013**, *49*, 1636–1638; (f) He, C.; Cao, W.; Zhang, J.; Ge, S.; Feng, X. Chiral *N,N*'-Dioxide/Scandium(III)-Catalyzed Asymmetric Alkylation of *N*-Unprotected 3-Substituted Oxindoles. *Adv. Synth. Catal.* **2018**, *360*, 4301–4305.

[14] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* allylation reactions, see: (a) Trost, B. M.; Frederiksen, M. U. Palladium-Catalyzed Asymmetric Allylation of Prochiral Nucleophiles: Synthesis of 3-Allyl-3-Aryl Oxindoles. *Angew. Chem. Int. Ed.* **2005***, 44,* 308–310; (b) Jiang, K.; Peng, J.; Cui, H.-L.; Chen, Y.-C. Organocatalytic asymmetric allylic alkylation of oxindoles with Morita-Baylis-Hillman carbonates. *Chem. Commun.* **2009**, 3955–3957; (c) Boucherif, A.; Duan, S.-W.; Yuan, Z.-G.; Lu, L.-Q.; Xiao, W.-J. Catalytic Asymmetric Allylation of 3-Aryloxindoles by Merging Palladium Catalysis and Asymmetric H-Bonding Catalysis. *Adv. Synth. Catal.* **2016**, *358,* 2594–2598; (d) Bleith, T.; Deng, Q.-H.; Wadepohl, H.; Gade, L. H. Radical Changes in Lewis Acid Catalysis: Matching Metal and Substrate. *Angew. Chem. Int. Ed.* **2016**, *55*, 7852–7856.

[15] Zhang, L.; Wang, M.; Zhou, M.; Zhang, Z.; Muraoka, M.; Zhang, W. Chiral Bicyclic Imidazole-Catalyzed Direct Enantioselective C-Acylation for the Synthesis of 2-Oxindoles Bearing a Quaternary Stereocenter. *Asian J. Org. Chem.* **2019**, *8*, 1024–1028.

[16] For selected reports on synthesis of 3,3-disubstituted oxindoles *via* arylation reactions, see: (a) Altman, R. A.; Hyde, A. M.; Huang, X.; Buchwald, S. L. Orthogonal Pd- and Cu-Based Catalyst Systems for C- and *N*-Arylation of Oxindoles. *J. Am. Chem. Soc.* **2008**, *130*, 9613–9620; (b) Guo, J.; Dong, S.; Zhang, Y.; Kuang, Y.; Liu, X.; Lin, L.; Feng, X. Chiral Scandium(III)-Catalyzed Enantioselective α-Arylation of N-Unprotected 3-Substituted Oxindoles with Diaryliodonium Salts. *Angew. Chem. Int. Ed.* **2013**, *52*, 10245–10249; (c) Shirakawa, S.; Koga, K.; Tokuda, T.; Yamamoto, K.; Maruoka, K. Catalytic Asymmetric Synthesis of 3,3'-Diaryloxindoles as Triarylmethanes with a Chiral All-Carbon Quaternary Center: Phase-Transfer-Catalyzed S_NAr Reaction. *Angew. Chem. Int. Ed.* **2014**, *53*, 6220–6223.

[17] Huang, J.; Bunel, E.; Faul, M. M. Palladium-Catalyzed α-Vinylation of Carbonyl Compounds. *Org. Lett.* **2007**, *9*, 4343–4346.

[18] Qiu, J.; Wu, D.; Karmaker, P. G.; Qi, G.; Chen, P.; Yin, H.; Chen, F.-X. Catalytic Asymmetric Electrophilic Cyanation of 3-Substituted Oxindoles. *Org. Lett.* **2017**, *19*, 4018–4021.

[19] Fu, L.; Greßies, S.; Chen, P.; Liu, G. Recent Advances and Perspectives in Transition Metal-Catalyzed 1,4-Functionalizations of Unactivated 1,3-Enynes for the Synthesis of Allenes. *Chin. J. Chem.* **2020**, *38*, 91–100.

[20] Zheng, Y.; Miao, B.; Qin, A.; Xiao, J.; Liu, Q.; Li, G.; Zhang, L.; Zhang, F.; Guo, Y.; Ma, S. Negishi Coupling for Highly Selective Syntheses of Allenes *via* Ligand Effect and Mechanistic Study *via* SAESI-MS/MS. *Chin. J. Chem.* **2019**, *37*, 1003–1008.

[21] Hoffmann-Röder, A.; Krause, N. Synthesis and Properties of Allenic Natural Products and Pharmaceuticals. *Angew. Chem. Int. Ed.* **2004**, *43*, 1196–1216.

[22] For reactions of allenes, see review: (a) Ma, S.; Hartung, J.; Kopf, T.; Murakami, M.; Matsuda, T.; Tius, M. A.; Zimmer, R.; Reissig, H.-U.; Hashmi, A. S. K.; Mandai, T.; Horváth, A.; Bäckvall, J.-E.; Krause, N.; Hoffmann-Röder, A.; Brummond, K. M.; Chen, H.; Wang, K. K. In *Modern Allene Chemistry*, Vol. 2, Wiley-VCH, Weinheim, **2004**, pp. 593–1123; (b) Zimmer, R.; Dinesh, C. U.; Nandanan, E.; Khan, F. A. Palladium-Catalyzed Reactions of Allenes. *Chem. Rev.* **2000**, *100,* 3067–3125; (c) Ma, S. Transition Metal-Catalyzed/Mediated Reaction of Allenes with a Nucleophilic Functionality Connected to the α-Carbon Atom. *Acc. Chem. Res.* **2003**, *36*, 701–712; (d) Ma, S. Some Typical Advances in the Synthetic Applications of Allenes. *Chem. Rev.* **2005**, *105*, 2829–2871; (e) Ma, S. Recent Advances in the Chemistry of Allenes. *Aldrichimica Acta* **2007***, 40*, 91–102; (f) Ma, S. Electrophilic Addition and Cyclization Reactions of Allenes. *Acc. Chem. Res.* **2009***, 42*, 1679–1688; (g) Aubert, C.; Fensterbank, L.; Garcia, P.; Malacria, M.; Simonneau, A. Transition Metal Catalyzed Cycloisomerizations of 1,*n*-Allenynes and -Allenenes. *Chem. Rev.* **2011**, *111*, 1954–1993; (h) Yu, S.; Ma, S. Allenes in Catalytic Asymmetric Synthesis and Natural Product Syntheses. *Angew. Chem. Int. Ed.* **2012**, *51*, 3074–3112; (i) Alcaide, B.; Almendros, P.; Aragoncillo, C. Cyclization reactions of bis(allenes) for the synthesis of polycarbo(hetero)cycles. *Chem. Soc. Rev.* **2014**, *43*, 3106–3135; (j) Muñoz, M. P. Silver and platinum-catalysed addition of O-H and N-H bonds to allenes. *Chem. Soc. Rev.* **2014**, *43*, 3164–3183; (k) Ye, J.; Ma, S. Palladium-Catalyzed Cyclization Reactions of Allenes in the Presence of Unsaturated Carbon-Carbon Bonds. *Acc. Chem. Res.* **2014**, *47*, 989–1000; (l) Liu, Y.; Bandini, M. Nickel Catalyzed Functionalization of Allenes. *Chin. J. Chem.* **2019**, *37*, 431–441.

[23] Han, Y.; Qin, A.; Ma, S. One Stone for Three Birds-Rhodium-Catalyzed Highly Diastereoselective Intramolecular [4+2] Cycloaddition of Optically Active Allene-1,3-dienes. *Chin. J. Chem.* **2019**, *37*, 486–496.

[24] (a) Bordwell, F. G. Equilibrium Acidities in Dimethyl Sulfoxide Solution. *Acc. Chem. Res.* **1988**, *21*, 456–463; (b) Uraguchi, D.; Torii, M.; Ooi, T. Acridinium Betaine as a Single-Electron-Transfer Catalyst: Design and Application to Dimerization of Oxindoles. *ACS Catal.* **2017**, *7*, 2765–2769; (c) Arnett, E. M.; Harrelson, Jr, J. A. A Spectacular Example of the Importance of Rotational Barriers: The Ionization of Meldrum's Acid. *J. Am. Chem. Soc.* **1987**, *109*, 809–812.

[25] Trost, B. M.; Van Vranken, D. L. Asymmetric Transition Metal-Catalyzed Allylic Alkylations. *Chem. Rev.* **1996**, *96*, 395−422.

[26] Braun, M. (*S*)-(-)-2-Allylcyclohexanone (2-Allylcyclohexan-1-one). *Org. Synth.* **2020**, *97*, 79–95.

[27] Gao, R.-D.; Zhai, Y.; You, S.-L.; Ma, S. Palladium-catalyzed intermolecular dearomatic allenylation of hydrocycloalk[*b*]indoles with 2,3-allenyl carbonates. *Org. Chem. Front.* **2018**, *5*, 1664–1669.

[28] For selected reports on the Pd-catalyzed racemic allenylation of 2,3-butadien-1-ol derivatives with malonates to provide allene

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product, see: (a) Djahanbini, D.; Cazes, B.; Gore, J. Reactive D'esters α-Alleniques. Synthese regiospecifique de diesters γ-alleniques et de dienes-1,3. *Tetrahedron Lett.* **1984**, *25*, 203–206; (b) Trost, B. M.; Tour, J. M. Synthesis of 4-(Dimethylphen-ylsilyl)buta-2,3-dien-1-ol and Its Use in Alkylation *J. Org. Chem.* **1989**, *54*, 484–486; (c) Wan, B.; Jia, G.; Ma, S. Palladium-Catalyzed Decarboxylation of Allenyl 3-Oxoalkanoates: An Efficient Synthesis of 3,4-Allenyl Ketones. *Org. Lett.* **2012**, *14*, 46–49.

[29] For selected reports on the Pd-catalyzed asymmetric allenylation of 2,3-butadien-1-ol derivatives with soft nucleophiles to provide allene product, see: (a) Imada, Y.; Ueno, K.; Kutsuwa, K.; Murahashi, S.-I. Palladium-Catalyzed Asymmetric Alkylation of 2,3-Alkadienyl Phosphates. Synthesis of Optically Active 2-(2,3-Alkadienyl)malonates. *Chem. Lett.* **2002**, 140–141; (b) Trost, B. M.; Fandrick, D. R.; Dinh, D. C. Dynamic Kinetic Asymmetric Allylic Alkylations of Allenes. *J. Am. Chem. Soc.* **2005**, *127*, 14186–14187; (c) Nemoto, T.; Kanematsu, M.; Tamura, S.; Hamada, Y. Palladium-Catalyzed Asymmetric Allylic Alkylation of 2,3-Allenyl Acetates Using a Chiral Diaminophosphine Oxide. *Adv. Synth. Catal*. **2009**, *351*, 1773–1778; (d) Li, Q.; Fu, C.; Ma, S. Catalytic Asymmetric Allenylation of Malonates with the Generation of Central Chirality. *Angew. Chem. Int. Ed.* **2012**, *51*, 11783–11786; (e) Boutier, A.; Kammerer-Pentier, C.; Krause, N.; Prestat, G.; Poli, G. Pd-Catalyzed Asymmetric Synthesis of *N*-Allenyl Amides and Their AuCatalyzed Cycloisomerizative Hydroalkylation: A New Route Toward Enantioenriched Pyrrolidones. *Chem.-Eur. J.* **2012**, *18*, 3840–3844; (f) Liu, H.-C.; Hu, Y.-Z.; Wang, Z.-F.; Tao, H.-Y.; Wang, C.-J. Synergistic Cu/Pd-Catalyzed Asymmetric Allenylic Alkylation of Azomethine Ylides for the Construction of α-Allene-Substituted Nonproteinogenic α-Amino Acids. *Chem.-Eur. J.* **2019**, *25*, 8681–8685; (g) Dai, J.; Duan, X.; Zhou, J.; Fu, C.; Ma, S. Catalytic Enantioselective Simultaneous Control of Axial Chirality and Central Chirality in Allenes. *Chin. J. Chem.* **2018**, *36*, 387–391; (h) Song, S.; Ma, S. Highly Selective Nucleophilic 4-Aryl-2,3-allenylation of Malonates. *Chin. J. Chem.* **2020**, *38*, 1233–1238.

[30] For selected reports on the Pd-catalyzed reaction of 2,3-butadien-1-ol derivatives with hard nucleophiles to provide 1,3-diene, see: (a) Kleijn, H.; Westmijze, H.; Meijer, J.; Vermeer, P. The palladium(0)-catalysed formation of 3-methoxy-1,3-butadienes from methoxypropadiene derivatives and organozinc compounds. *Recl. Trav. Chim. Pays-Bas.* **1983**, *102*, 378–380; (b) Djahanbini, D.; Cazes, B.; Gore, J. Synthèse de Triméthylsilylméthyl-2 dièhes-1,3. *Tetrahedron* **1985**, *41*, 867–873; (c) Ni, Z.; Padwa, A. Palladium Catalyzed Cross Coupling Reaction of α-Allenyl Acetates under Nonbasic Conditions. *Synlett* **1992**, 869–870; (d) Moriya, T.; Furuuchi, T.; Miyaura, N. A New Facile Synthesis of 2-Substituted 1,3-Butadiene Derivatives *via* Palladium-Catalyzed Cross-Coupling Reaction of 2,3-Alkadienyl Carbonates with Organoboron Compounds. *Tetrahedron* **1994**, *50*, 7961–7968; (e) Lin, J.; Zhu, T.; Jia, M.; Ma, S. A Pd-catalyzed ring opening coupling reaction of 2,3-allenylic carbonates with cyclopropanols. *Chem. Commun.* **2019**, *55*, 4523–4526; (f) Hu, S.; Sun, W.; Chen, J.; Li, S.; Zhao, R.; Xu, P.; Gao, Y.; Zhao, Y. Palladium-catalyzed C-P cross-coupling of allenic alcohols with H-phosphonates leading to 2-phosphinoyl-1,3-butadienes. *Chem. Commun.* **2021**, *57*, 339–342.

[31] For detection of α-methylene-π-allylic Pd intermediate, see: Ogasawara, M.; Okada, A.; Watanabe, S.; Fan, L.; Uetake, K.; Nakajima, K.; Takahashi, T. Synthesis, Structure, and Reactivity of (1,2,3-η^3-Butadien-3-yl)palladium Complexes. *Organometallics* **2007**, *26*, 5025–5029.

[32] For all-carbon 3,3-rearrangement-type reductive elimination of bis(allyl)metal species, see: (a) Méndez, M.; Cuerva, J. M.; Gómez-Bengoa, E.; Cárdenas, D. J.; Echavarren, A. M. Intramolecular Coupling of Allyl Carboxylates with Allyl Stannanes and Allyl Silanes: A New Type of Reductive Elimination Reaction. *Chem.-Eur. J.* **2002**, *8*, 3620–3628; (b) Zhang, P.; Brozek, L. A.; Morken, J. P. Pd-Catalyzed Enantioselective Allyl-Allyl Cross-Coupling. *J. Am. Chem. Soc.* **2010**, *132*, 10686–10688.

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